CONSTRUCTAI: AN AI-DRIVEN PRE-CONSTRUCTION FEASIBILITY ANALYZER FOR INTEGRATED CONSTRUCTION DECISION SUPPORT

COMMON INTEGRATED SOLUTION REPORT

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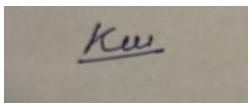
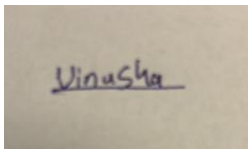
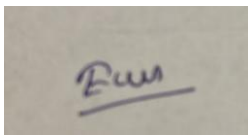
Sri Lanka

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ABSTRACT

Pre-construction planning in Sri Lanka is divided among four professionals who each produce a document and hand it on. None of their analyses is evaluated against the others, so site-driven risk, measurement error, regulatory exposure and long-run operating cost are each discovered after the decisions that caused them have been taken. The industry pays for this in budget overruns and in buildings that fail earlier than their owners expect.

ConstructAI performs all four analyses over a single shared project record. Component 1 derives six environmental exposure parameters and terrain buildability from open satellite, elevation, soil and climate data. Component 2 extracts metric quantities and six structural parameters from ordinary 2D floor plans using a tiled YOLOv8s detector with OCR-based scale calibration. Component 3 evaluates the design against a codified subset of the Urban Development Authority regulations using a forward-chaining rule engine. Component 4 predicts structural service life with an XGBoost regressor, prices the bill of quantities at live rates and computes a discounted fifty-year trajectory. The components communicate only through a provenance-carrying shared state in which a missing value is explicitly unknown rather than silently defaulted.

The integrated model reached a coefficient of determination of 0.9847 and a root mean squared error of 0.433 years on a held-out partition of a 1 000-record physics-informed dataset. The decisive result is that integration changes the advice instead of merely collecting it: holding the building constant, site conditions alone move predicted service life by 6.40 years across four realistic locations, a 36.1 percent reduction. Two uncomfortable findings are also reported. Eight of the thirteen parameters carry essentially no information, and of the five that do, the two largest are user judgements no design document can supply. The training data is synthetic, so these figures characterise the formulation rather than real buildings.

Keywords: integrated decision support, pre-construction feasibility, geospatial analysis, computer vision, automated compliance checking, lifecycle cost prediction

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AGPL	Affero General Public License
API	Application Programming Interface
BIM	Building Information Modelling
BOQ	Bill of Quantities
CAPEX	Capital Expenditure
CPU	Central Processing Unit
CV	Cross-Validation
DEM	Digital Elevation Model
FAR	Floor Area Ratio
GDAL	Geospatial Data Abstraction Library
GIS	Geographic Information System
IoU	Intersection over Union
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
mAP	Mean Average Precision
MSL	Mean Sea Level
NFR	Non-Functional Requirement
OCR	Optical Character Recognition
ODbL	Open Database License
OPEX	Operational Expenditure
OSM	OpenStreetMap
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology

Abbreviation	Meaning
SRTM	Shuttle Radar Topography Mission
UAT	User Acceptance Testing
UDA	Urban Development Authority (Sri Lanka)
WCAG	Web Content Accessibility Guidelines
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once (object detection family)

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Construction is the largest single financial commitment most Sri Lankan households ever make, and the phase that determines how much of that money is well spent lasts only a few weeks. Once a plot has been bought and a design frozen, the cost envelope of the project is essentially fixed; everything after that is execution against a decision taken on incomplete information.

It is incomplete for a structural reason, not a careless one. Pre-construction work is divided among four professionals who each produce a document and hand it on. A land surveyor assesses the plot. An architect prepares drawings. A lawyer verifies the title and the zoning position. A quantity surveyor measures the drawings and produces a bill of quantities. Every one of those is competent work performed on a private copy of the truth, and the handover artefact is a document. Documents do not recompute. When the surveyor notes that the rear third of the plot sits below the flood line, nothing in the architect's massing changes; when the architect specifies mild steel reinforcement for a site three hundred metres from the sea, nothing in anybody's maintenance forecast changes.

Industry pays for this in overruns. Across a large international sample of projects, roughly nine in ten exceeded their approved budget with mean overruns near 28 percent, a pattern too consistent across decades and countries to be explained by ordinary estimating noise [1]. Smaller residential projects are not exempt and are arguably worse served, because the client frequently cannot afford a full professional team and the takeoff is done quickly and by hand, with error rates in the 10 to 15 percent range commonly reported for early-stage estimates produced under time pressure [2].

Sri Lanka adds two local aggravating factors. The island's wet zone sits on highly weathered lateritic soils whose reaction is naturally acidic, and a large share of its population and its construction activity sits within a few kilometres of a coastline where airborne chloride is a continuous load on reinforced concrete. Both conditions

shorten the service life of a building in ways that are well documented in the durability literature [4], [5] and are almost never assessed before a plot is bought.

1.2 Stakeholders and user needs

Table 1.1 records the stakeholders identified during the proposal stage and the needs confirmed in discussion with practitioners.

Table 1.1: Stakeholders and confirmed needs

Stakeholder	Relationship to the problem	Confirmed need
Individual land buyer / homeowner	Makes the purchase decision with the least information of anyone involved	To know, before paying, what the land and the design commit them to over the building's life
Small and medium contractor or architectural practice	Produces early-stage estimates under time pressure	A fast, defensible quantity takeoff and a specification recommendation they can show a client
Quantity surveyor	Measures drawings by hand; absorbs the error when rushed	A first-pass takeoff to check rather than to replace their judgement
Local authority / UDA planning officer	Receives submissions that are non-compliant and must be returned	Fewer submissions that fail on basic setback, coverage or plot-size rules
Bank or insurer	Holds construction loan and property collateral risk	A quantified, repeatable site and structure risk profile per property

One need recurred in every conversation and shaped the project more than any other, was not for a better cost estimate. It was for the cost estimate to include the part that comes after handover. Practitioners were clear that clients ask what the building will cost to build and almost never ask what it will cost to keep.

1.3 Alignment with the Sustainable Development Goals

Three goals are the relevant ones, and we have tried to state the alignment at the size it actually is.

- **SDG 9, Industry, Innovation and Infrastructure**, target 9.4 on upgrading infrastructure and retrofitting industries to make them sustainable. Automated takeoff and specification advice reduce over-ordering of cement and steel, which is a significant and largely invisible source of waste on small sites.
- **SDG 11, Sustainable Cities and Communities**, targets 11.1 on adequate, safe and affordable housing and 11.5 on reducing losses from disasters. A buyer

warned before purchase that a plot lies in a drainage line or a high-salinity band can avoid it or build for it.

- **SDG 12, Responsible Consumption and Production**, target 12.5 on reducing waste generation. Premature structural failure means demolition and rebuilding, and cement is among the most carbon-intensive materials in common use.

What we have not done is measure any of this. No part of the project quantified avoided emissions, avoided waste or avoided housing loss. These are plausible contributions from the mechanism of the solution, not demonstrated outcomes, and we have deliberately not attached numbers to them.

1.4 Purpose and scope of the common report

This report presents ConstructAI as one solution rather than as four projects. It covers the shared problem, the architecture, how the four components interoperate, how the integrated system was built and tested, what the integrated evaluation found, and what the commercial and sustainability case looks like.

It does not repeat the component-level research. Each of the four individual theses contains the detailed literature review, methodology, design, implementation and evaluation for its own component, and this report cites them instead of reproducing them. Where a result in Chapter 6 was produced by one member's analysis, the attribution is stated.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem. Pre-construction decisions in Sri Lanka are made from fragmented, manually produced documents that are never evaluated against one another, so that site-driven risk, measurement error, regulatory exposure and long-run operating cost are each discovered too late to be acted on cheaply.

That statement is about the decision process, not about any technology. It is worth separating the two because the obvious technological framings, that the industry needs BIM, or needs machine learning, or needs a mobile app, all presuppose a particular answer. The problem as we have defined it would exist even if every one of those were adopted, as long as the four analyses remained unconnected.

2.2 Root causes and overall research gap

Four causes sustain the problem.

1. **Professional fragmentation.** The four pre-construction analyses are performed by different professionals at different times with no shared artefact. Each handover loses the ability to recompute.
2. **Data inaccessibility.** The information needed to assess site risk exists in open global datasets but requires geospatial skills the buyer, the mason and often the architect do not have.
3. **Manual measurement.** Quantity takeoff from 2D drawings is slow and error-prone, and building information modelling, which would solve it, presumes an adoption step the small-project market has not taken.
4. **Capital-only costing.** Traditional bills of quantities compute initial capital expenditure and stop. A building that is cheaper to build and needs its roof replaced twice as often is recorded as the cheaper building.

What follows is a gap of integration, not of capability. Each subproblem has a respectable literature. Geographic information systems have been used for construction suitability mapping for decades and more recently with satellite imagery

[3]. Deep object detection has been applied to floor plan parsing with strong results [7], [8]. Rule-based and semantic engines for automated building code compliance are well developed [9], [10]. Gradient boosting is the established approach for tabular construction cost prediction [11], [12]. What does not exist, as far as our review found, is a platform that takes a map pin and a floor plan and returns, in one pass, a terrain and environmental risk assessment, a measured bill of quantities, a statutory approval roadmap and a fifty-year cost trajectory that reflects the specific materials chosen for that specific site.

2.3 Subproblem and component mapping

Our four subproblems do not overlap in what they measure and are interdependent in what they need. Table 2.1 gives the mapping.

Table 2.1: Subproblem to component mapping

Subproblem	Component and owner	Input	Core output
Site risk is invisible before purchase	C1 Automated Site Analysis — Ranaweera K. A.	Coordinate, plot polygon	Six exposure parameters, slope banding, flood flag
Quantities are locked in drawings	C2 Blueprint Extractor — Shavinka Y. G. K.	Floor plan PDF or image	Metric quantity table, six structural parameters
Regulatory position is discovered late	C3 Compliance Advisory — Samuditha T. A. V.	Geometry, zone, frontage	Violations, approval roadmap, permit fee estimate
Only capital cost is computed	C4 Cost and Lifecycle — Shamindi E. E.	All of the above, live prices	Capital cost, predicted service life, 50-year trajectory

Why all four are necessary is easiest to see by removing one. Without C1 the cost model cannot distinguish a coastal plot from an inland one, and the sensitivity results in Section 6.3 show that distinction is worth 6.4 years of service life. Without C2 there are no quantities, so there is no capital cost and no structural parameter vector. Without C3 the design may be unbuildable as drawn, which invalidates every quantity downstream of it. Without C4 the other three produce three disconnected reports, which is the situation the project exists to replace.

2.4 Overall aim and group objectives

Aim. To design, build and evaluate an integrated platform that performs the four principal pre-construction analyses over a single shared project record, so that the output of each analysis constrains the others, and to demonstrate that the integration produces decision-relevant advice that none of the four analyses could produce alone.

Table 2.2 states the aim as six measurable group objectives, maps each one to the components that implement it, and names the evidence used to judge whether it was met.

Table 2.2: Group objectives, component mapping and evidence

ID	Group objective	Components	Evidence
GO1	Derive quantified site exposure and terrain constraints from open data for any Sri Lankan coordinate	C1	Section 6.3, C1 thesis Ch. 6
GO2	Extract metrically correct quantities and structural parameters from ordinary 2D drawings	C2	Section 6.2, C2 thesis Ch. 6
GO3	Codify a working subset of the UDA Planning and Development Regulations into a machine-checkable rule set	C3	Section 6.2, C3 thesis
GO4	Predict structural service life and a fifty-year cost trajectory from the combined parameters	C4	Section 6.3, C4 thesis
GO5	Operate the four components over one shared project state with no manual data transfer	All	Sections 4.2, 6.2
GO6	Demonstrate that integration changes the advice given, not merely the number of reports produced	All	Sections 6.3.4, 6.5

Table 2.3 traces each group objective to the component requirements that implement it and to the evidence that demonstrates it, so that no objective is left without a verifiable outcome.

Table 2.3: Requirement traceability from group objective to evidence

Objective	Implementing requirement	Owner	Verification method	Evidence
GO1	C1 FR1–FR8: derive exposure parameters, slope banding and flood flag	C1	Reference agreement against QGIS and published station values	C1 thesis 6.2
GO2	C2 FR1–FR9: normalise, detect, calibrate scale, synthesise quantities	C2	Detection metrics and comparison against independent manual takeoff	C2 thesis 6.2
GO3	C3 rule families: density,	C3	Replay of submissions	C3 thesis

Objective	Implementing requirement	Owner	Verification method	Evidence
	dimensional, spatial, procedural		with recorded municipal outcomes	
GO4	C4: feature assembly, service life regression, discounted lifecycle costing	C4	Held-out partition with cross-validation against five baselines	Section 6.3.1
GO5	Shared project state contract; range assertions; null semantics	All	Interface and integration tests; fault injection	Section 6.2
GO6	Cross-component scenario evaluation	All	Scenario comparison with the structure held constant	Section 6.3.4

2.5 Scope assumptions and constraints

We target low-rise residential and light-commercial buildings in Sri Lanka. High-rise, industrial and marine structures are out of scope: their exposure conditions, foundation requirements and regulatory treatment are different enough that the calibration would not transfer.

Four assumptions constrain the whole solution and are stated here and not buried in the component theses.

- A plot is adequately represented by a single centroid coordinate for climate and proximity purposes. Reasonable for a 10 to 20 perch residential plot; not reasonable for an estate.
- A design drawing is available as a PDF or an image drawn to a consistent stated scale. The platform does not require BIM and cannot exploit it.
- The regulatory rule set reflects the Urban Development Authority regulations and municipal schedules as at the stated effective date [13]. Regulations change by gazette; a stale rule store gives confidently wrong advice.
- Service life prediction is trained on a physics-informed synthetic dataset rather than on observed buildings. Section 6.6 treats this as the principal threat to the validity of the whole project.

That dependency structure also constrains operation: C4 cannot run until C1 and C2 have both written to the project state, and C3's verdict can invalidate a geometry that C2 has already measured. Section 4.4 describes how the orchestration handles that.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

ConstructAI is one system with four analysis engines over a single shared project state, not four tools behind one login. The distinction is concrete rather than rhetorical: the engines do not call each other, they read and write a common record, and it is that record which makes a finding in one analysis change a result in another. When C1 writes a coastal-proximity value into the record, C4 recomputes the predicted service life without any explicit call between them.

At the centre of it, a buyer gets, before paying for the land, the four answers that currently arrive separately and late: can this land be built on, what will the building take in materials, will the authority approve it as drawn, and what will it cost over fifty years, not just over the construction programme.

3.2 High level architecture

Figure 3.1 shows the platform as a whole: the two user inputs, the four analysis engines, the shared project state that mediates every exchange between them, and the integrated report that is assembled once all four have run.

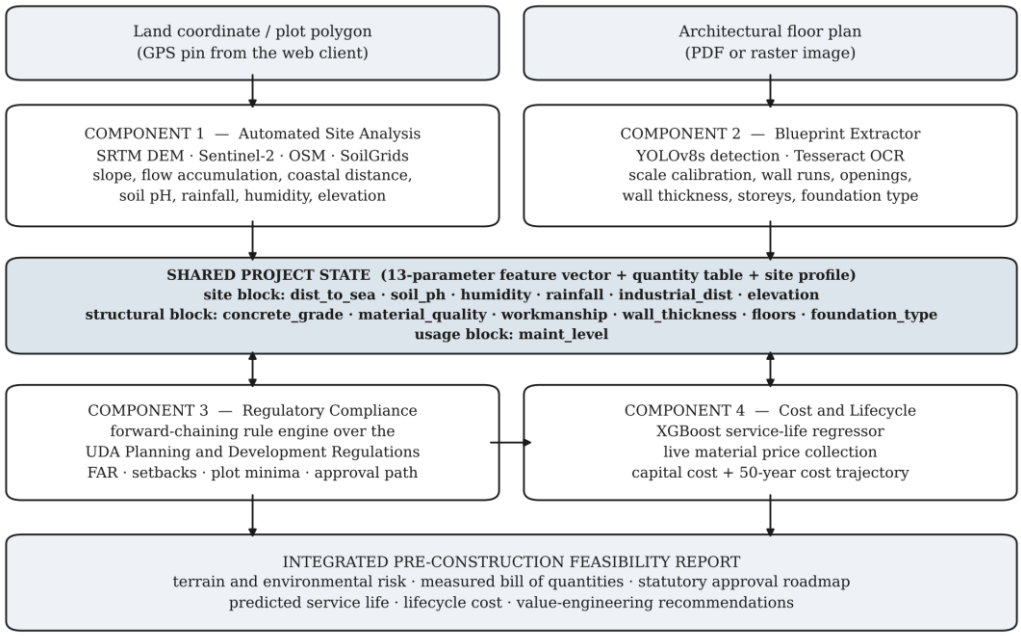


Figure 3.1: High level architecture of the ConstructAI platform.

Two inputs enter the system: a land coordinate with an optional plot polygon, and an architectural floor plan. C1 and C2 are independent and run in parallel; both write into the shared project state. C3 and C4 read from it, and C3's compliance constraints feed C4 before the cost and lifecycle computation. The integrated report is assembled from the state once all four have completed or reported a failure.

Shared project state is a JSON document with four groups of fields. The site block holds the six parameters C1 derives. The geometry block holds the quantity table and the six structural parameters C2 derives. The regulatory block holds the zone, permitted floor area ratio, setback requirements and detected violations from C3. The specification block holds the assembled thirteen-element feature vector that C4 consumes, together with pricing and lifecycle outputs. Every field carries provenance: which component wrote it, from which source, and when.

Everything deploys as a single web application with the four engines as stateless services behind it. Statelessness was a deliberate early decision; it means the engines scale horizontally without coordination and, more importantly during development, that each could be tested in isolation against a fixture.

3.3 Component 1 summary

Table 3.1 summarises Component 1 and Figure 3.2 shows its internal pipeline. The detailed method and evaluation are in that member's individual thesis.

Table 3.1: Component 1 summary

Attribute	Detail
Owner	Ranaweera K. A. (IT22542724)
Subproblem	Site and environmental risk is invisible to a buyer before purchase
Input	WGS84 coordinate and optional plot polygon
Core method	Raster and vector derivation over SRTM elevation [14], Sentinel-2 imagery [15], OpenStreetMap vectors, SoilGrids soil predictions [16] and Department of Meteorology normals, using Horn slope estimation [17], D8 flow accumulation [18] and geodesic proximity analysis
Output	dist_to_sea, soil_ph, humidity, rainfall, industrial_dist, elevation, plus mean slope, buildable fraction and a flood-proximity flag
Contribution to the whole	Supplies six of the thirteen model features and the terrain basis for C3's coverage and setback checks; 17.4 percent of total model gain

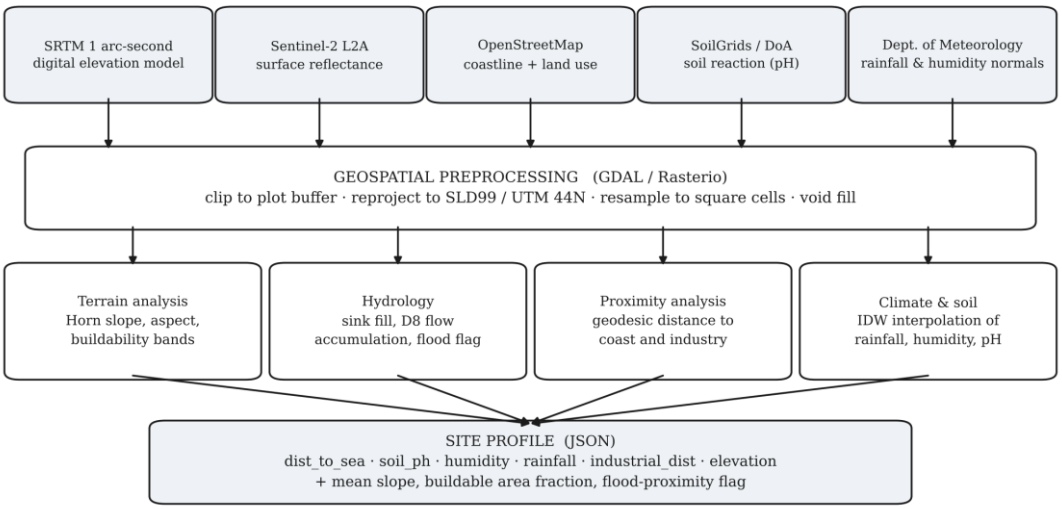


Figure 3.2: Component 1 processing pipeline.

3.4 Component 2 summary

Table 3.2 summarises Component 2 and Figure 3.3 shows its internal pipeline, using the same headings as Component 1 so that the two can be compared and the division of work is visible.

Table 3.2: Component 2 summary

Attribute	Detail
Owner	Shavinka Y. G. K. (IT22110770)
Subproblem	Structural quantities and parameters are locked inside 2D drawings
Input	Architectural floor plan as a vector PDF or raster image
Core method	Adaptive thresholding and Hough-based deskew [19], tiled YOLOv8s detection of walls, columns, doors and windows [6], [7], Tesseract OCR [20] with a median metres-per-pixel scale estimator, wall-run merging and clash detection
Output	Metric quantity table (wall length and area, opening schedule, gross floor area, clash report) and six structural parameters
Contribution to the whole	Supplies six of the thirteen model features and the entire capital cost basis; 82.6 percent of total model gain

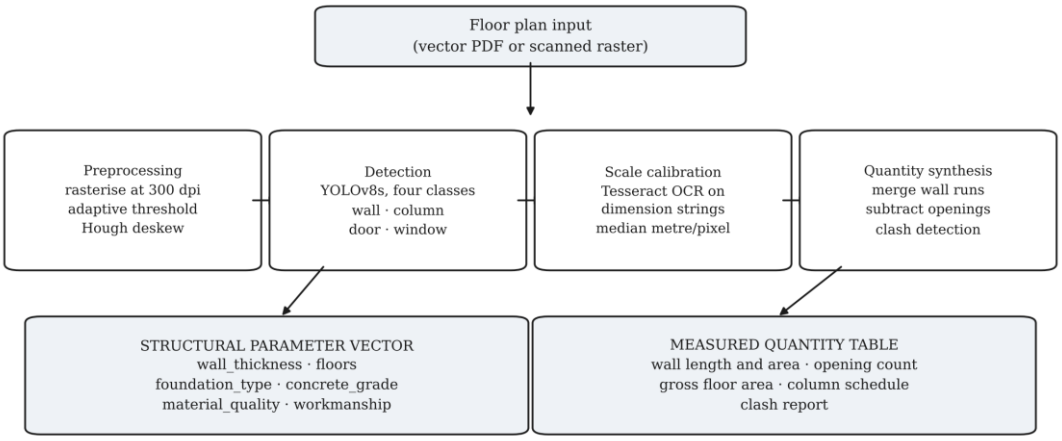


Figure 3.3: Component 2 processing pipeline.

3.5 Component 3 summary

Table 3.3 summarises Component 3 under the same headings.

Table 3.3: Component 3 summary

Attribute	Detail
Owner	Samuditha T. A. V. (IT22542892)
Subproblem	Regulatory non-compliance is discovered after design, when correction is expensive
Input	Gross floor area, storey count, plot geometry, zone and road frontage from the shared state
Core method	Forward-chaining rule engine over a declarative encoding of the UDA Planning and Development Regulations and municipal schedules [13], following the rule interpretation / model preparation / execution / reporting decomposition of Eastman et al. [9]
Output	Floor area ratio, setback and plot-minimum verdicts with clause citations, an approval roadmap and a permit fee estimate
Contribution to the whole	Constrains the buildable geometry that C2 measured and C4 prices; prevents a cost estimate for a design that cannot be approved

3.6 Component 4 summary

Table 3.4 summarises Component 4, the last engine in the chain and the one whose output the user acts on.

Table 3.4: Component 4 summary

Attribute	Detail
Owner	Shamindi E. E. (IT22547460)
Subproblem	Costing stops at capital expenditure and ignores the building's operating life

Attribute	Detail
Input	The thirteen-element feature vector assembled from C1, C2 and the user, plus the quantity table and live material prices
Core method	XGBoost regression [21] for service life prediction, scheduled web scraping for material prices, and discounted lifecycle costing following ISO 15686-5 [22]
Output	Capital cost, predicted service life, fifty-year discounted cost trajectory and value-engineering recommendations
Contribution to the whole	The decision hub; converts the other three components' measurements into the advice the user acts on

3.7 End to end user and data flow

A single pass through the system looks like this, and Figure 3.4 shows the same sequence as a data flow.

5. The user drops a pin on a plot and uploads a floor plan.
6. C1 retrieves and processes the elevation, imagery, soil and climate layers for that location and writes the site block: six exposure parameters, mean slope, buildable fraction and a flood-proximity flag.
7. In parallel, C2 normalises the drawing, detects the structural elements, recovers the scale from the drawing's own dimension text and writes the geometry block: the quantity table and six structural parameters.
8. C3 reads the gross floor area, storey count and plot geometry, evaluates them against the rule store for the applicable zone and frontage, and writes verdicts, clause citations and an approval roadmap into the regulatory block.
9. C4 assembles the thirteen-element feature vector from the site block, the geometry block and the user-supplied maintenance level, prices the quantity table at current scraped material rates, predicts service life and computes the discounted fifty-year trajectory.
10. The platform assembles the integrated report and presents an alternative-specification comparison, in which the same building is costed under two or more material specifications and the lifecycle consequence of each is shown.

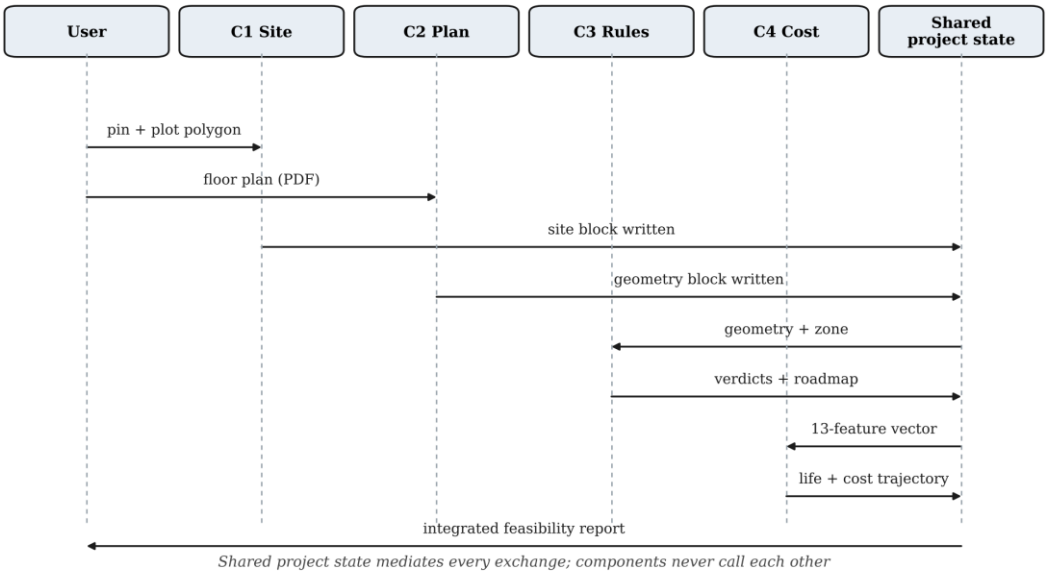


Figure 3.4: End-to-end data flow for a single analysis.

Integration earns its keep at the points where components touch each other. C1's coastal value changes C4's prediction. C2's gross floor area changes C3's floor area ratio verdict, and a C3 violation changes the geometry C2's quantities describe. C3's buildable envelope changes what C4 is allowed to price. None of these would happen in the document-based workflow the platform replaces.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

We adopted an interface-first strategy in the second sprint, before any component worked end to end. We agreed the shape of the shared project state, committed a fixture file containing a realistic site profile, a realistic quantity table and a realistic compliance verdict, and everybody built against the fixture. Components were then integrated one at a time against the real producers, in the order C1, C2, C3, C4, so that each integration step introduced exactly one new source of error.

This is the single decision that most affected how the project went, and it is worth being honest about why we made it. It was not foresight. It was that two members were blocked for a fortnight in the first sprint waiting for upstream output that did not exist yet, and the fixture was the obvious way to stop that happening again.

Interface ownership was explicit: the producing component owns the schema of the block it writes, and any change to that schema requires a pull request that the consuming members review. Three schema changes went through that process during the project, all of them in the first half.

4.2 Interfaces data contracts and shared models

Shared project state is the only interface. Table 4.1 gives the contract.

Table 4.1: Shared project state contract

Block	Fields	Written by	Read by
site	dist_to_sea (m), soil_ph, humidity (%), rainfall (mm/yr), industrial_dist (m), elevation (m MSL), mean_slope (deg), buildable_fraction, flood_proximity_flag	C1	C3, C4, report
geometry	wall_thickness (in), floors, foundation_type, concrete_grade (MPa), material_quality, workmanship, wall_length_m, wall_area_m2, gross_floor_area_m2, opening_schedule, clash_report, scale_source	C2	C3, C4, report
regulatory	zone, far_permitted, far_actual, setback_requirements, violations[], approval_roadmap, permit_fee_estimate, rule_effective_date	C3	C4, report
specification	feature_vector (13 keyed values), capital_cost, predicted_service_life_years,	C4	report

Block	Fields	Written by	Read by
	lifecycle_trajectory, recommendations[]		
provenance	per-field source, retrieval or derivation timestamp, confidence where applicable	All	Audit, report

Four conventions make the contract work and all four were learned the hard way.

- **Units live in the field name.** An early integration failure had C1 emitting metres while C4's feature assembly assumed kilometres, and another had C2 emitting inches while the assembly assumed millimetres. Both produced predictions that looked plausible and were wrong. Suffixing the unit, and asserting ranges at the boundary, ended that class of bug.
- **The feature vector is keyed, not positional.** The first version was a list and we silently transposed two columns.
- **Null means unknown, never a default.** A failed data source degrades one field and the consuming component treats it as missing. An early version imputed a neutral soil pH of 7.0 on failure, which quietly removed an entire risk category from the prediction.
- **Every field carries provenance.** A value derived from a 250 m soil grid and a value from a laboratory report are not the same kind of thing and the interface does not pretend they are.

4.3 Shared nonfunctional requirements

Table 4.2 lists the non-functional requirements that apply across the whole solution rather than to any single component, together with the evidence used to demonstrate each one.

Table 4.2: Shared non-functional requirements across the integrated solution

Quality	Requirement	Target	How it is demonstrated
Performance	End-to-end response for a single-plot, single-sheet analysis	Under 60 s warm	Section 6.2 timing
Reliability	A failed data source or engine degrades the report, never the request	No request failures from one source outage	Fault injection, Section 6.2
Security	Authentication on saved projects; no third-party transmission of user data	No plan or coordinate leaves	Code review and log inspection

Quality	Requirement	Target	How it is demonstrated
		the platform	
Privacy	Coordinates and drawings not retained unless the project is saved	Non-retention by default	Log inspection, Section 7.4
Usability	A non-specialist can complete the main workflow unaided	Task completion without assistance	UAT, Section 6.4
Transparency	Every output value traceable to its source and confidence	100 percent of fields carry provenance	Interface contract, Table 4.1
Maintainability	Regulation changes are data edits, not code changes	Rule store editable without redeployment	C3 design
Accessibility	Readable contrast, keyboard navigation, no colour-only encoding of risk	WCAG 2.1 AA for the report view	Manual audit of the report view against the AA criteria

Transparency deserves a word, because it is unusual to treat as non-functional. The platform's headline output is a number of years, and a number of years with no visible provenance invites a user to trust it further than the evidence supports. Making provenance a hard requirement across all four components was a deliberate counterweight to that.

4.4 Integrated workflow and exception handling

Normal operation is the sequence in Section 3.7. The orchestration layer runs C1 and C2 concurrently, waits for both, runs C3, then C4. The important behaviour is what happens when something goes wrong, and the group spent more design effort on that than on the happy path.

Table 4.3: Failure paths and system behaviour

Failure	Behaviour	What the user sees
One C1 data source unreachable	That field is null; the rest of the site block is written; C4 treats the feature as missing	The affected risk factor is shown as not assessed, with the reason
C2 cannot establish drawing scale	No quantities are emitted at all (C2 FR6)	A request for a reference line, or a clear statement that quantities cannot be produced
C2 detection confidence very low across a sheet	Quantities are emitted but flagged	Quantities marked low confidence and recommended for manual check
C3 cannot evaluate a rule for want of an input	The rule reports unknown, never pass	The rule is listed as not assessed, with what is missing

Failure	Behaviour	What the user sees
Material price source stale beyond the window	Last known price used and flagged	Capital cost marked as using prices from a stated date
C4 feature vector fails a range assertion	The prediction is withheld	An explicit error rather than a plausible wrong number

One principle sits behind every row: the system is allowed to say it does not know, and it is not allowed to say something it cannot support. A compliance tool that reports silence as approval, or a cost tool that defaults a missing parameter, is a liability instead of a feature.

4.5 Team contribution and dependency matrix

Table 4.4 maps each member to their component, to the research and development tasks they carried out, and to the shared and integration work.

Table 4.4: Team contribution and dependency matrix

Member	Component	Research tasks	Development tasks	Shared and integration work
Ranaweera K. A. (IT22542724)	C1	Exposure parameter selection; open data source survey; sensitivity analysis of the site block	Raster access layer; terrain and hydrology units; proximity and climate units	Site profile contract; environmental terms of the degradation logic; joint re-run of the model evaluation with a held-out partition
Shavinka Y. G. K. (IT22110770)	C2	Class selection; detector evaluation protocol; recoverability analysis of structural parameters	Preprocessing; tiled detection; OCR scale calibration; quantity synthesis and clash check	Geometry contract; structural parameter definitions for the degradation logic
Samuditha T. A. V. (IT22542892)	C3	Regulation review and rule formalisation; validation against recorded decisions	Forward-chaining rule engine; rule store; approval roadmap and fee logic	Regulatory contract; buildable envelope constraints passed to C4
Shamindi E. E. (IT22547460)	C4	Degradation function; model selection, tuning and benchmarking; lifecycle cost formulation; value-engineering and heuristic comparison	Dataset generation; feature assembly; XGBoost training and serving; price scraper; discounting; advisory layer	Specification contract; joint re-run of the model evaluation with a held-out partition; integrated report assembly
All members	—	Problem definition; stakeholder discussions; shared NFR definition	Fixture file; range assertions; orchestration layer	Interface-first integration; end-to-end testing; UAT sessions

Appendix B carries the signed contribution declaration that corresponds to Table 4.4. The two things worth noting here are that the degradation logic used to generate the training dataset was written jointly, with each member contributing the terms for their own domain, and that the integration layer was genuinely shared work and not assigned to one member, which is the main reason the interface problems in Section 4.2 were caught at all.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

We standardised on Python 3.11 across all four components. That choice was made in the first sprint for a practical reason instead of a principled one: it meant the shared project state could move between components as a plain dictionary during development, without a serialisation boundary to debug, and it meant one virtual environment and one requirements file for everybody.

Table 5.1 lists the shared stack and what each layer is for.

Table 5.1: Shared technology stack

Layer	Technology	Used by	Reason
Language	Python 3.11	All	One environment; dictionaries across component boundaries during development
Geospatial	GDAL / Rasterio, Shapely, pyproj	C1	Reference implementations for raster access and geodesic geometry
Vision	OpenCV, Ultralytics YOLOv8s, Tesseract	C2	Mature detection and OCR tooling with acceptable CPU latency
Rules	Custom forward-chaining evaluator over a JSON rule store	C3	Auditable, and a regulation change is a data edit not a code change
Machine learning	XGBoost [21], scikit-learn [23], Pandas, NumPy, joblib	C4	Established for tabular regression; fast enough to run inside a request
Data collection	BeautifulSoup	C4	Scheduled material price collection
Web	Web application front end over stateless component services	All	Horizontal scaling without coordination
Testing	pytest, Selenium	All	Unit and component tests; automated UI regression
Collaboration	Git / GitHub, MS Teams, MS Planner	All	Branch-per-component with pull request review; traceable task and meeting records

5.2 Development and integration process

Work was organised in two-week sprints with a planning session at the start and a demonstration at the end. The practice that mattered most was interface-first development, described in Section 4.1: the shared state schema and a fixture file

were agreed before any component produced real output, and every member developed against the fixture until their upstream producer was ready.

Version control used a branch per component with pull requests reviewed by at least one other member. The review requirement was occasionally slow and caught two of the three interface defects described in Section 4.2, which settled the argument about whether it was worth keeping.

Issue management used GitHub issues for technical work and MS Planner for the project-level tasks that were not code, principally the regulation review, the drawing collection and the acceptance sessions. Supervisor meetings were recorded in MS Teams with decisions written down, which turned out to matter more than expected when the group revisited why the depreciation heuristic had been set at five percent.

5.2.1 Sprint schedule and milestones

Table 5.2 records the sprint structure and what each one delivered. The column that matters is the last one: what actually shipped, as distinct from what was planned.

Table 5.2: Sprint structure and focus

Sprint	Focus	Principal outcome
S1	Problem definition, stakeholder discussions, component scoping	Scope and the four-component split agreed; first data source surveys
S2	Shared state schema; fixture file; first vertical slice	Schema and fixture agreed, which unblocked every member working downstream
S3	Component 1 raster pipeline; Component 2 preprocessing and first detector run	Both producers emitting their blocks against the agreed schema
S4	Component 3 rule formalisation; Component 4 dataset generation and first model	Rule store populated with the four families; first trained service life model
S5	Producer-to-consumer integration; unit encoding and range assertions	The unit and ordering defects described in Section 4.2 found and fixed
S6	Full orchestration; failure paths; progress presentation	End-to-end run with every failure path in Table 4.3 implemented
S7	Model re-evaluation with a held-out partition; sensitivity and scenario studies	The results reported in Section 6.3, replacing a full-data fit
S8	User acceptance sessions; report writing; final hardening	Break-even presentation adopted; documentation completed

5.2.2 Risk management

Risks were reviewed at each sprint planning session. Table 5.3 lists those that materialised or came close, with what was actually done and not what was planned.

Table 5.3: Project risk register and how each risk was handled

Risk	Impact	Response taken	Outcome
No real building dataset available for training or validation	High	Generated a physics-informed dataset from published relationships and an industry heuristic; disclosed the consequence throughout	Materialised; bounds every quantitative claim (Section 6.6.1)
Members blocked waiting on upstream components	High	Interface-first development against a committed fixture	Materialised in the first sprint; resolved from the second
Silent unit mismatches across component boundaries	High	Units encoded in field names; range assertions at every boundary	Materialised twice; both caught in pull-request review
Insufficient drawings for detector training	High	Pre-training on CubiCasa5K [8]; targeted local collection under permission	Managed; drawing-style drift remains a live limitation (Section 6.6.4)
Regulation changes during the project	Medium	Effective date stored with every rule and surfaced with the verdict	Did not materialise within the project period
Upstream data source outage or version change	Medium	Null-means-unknown semantics; version pinning; scheduled drift check	Exercised deliberately by fault injection
Licensing constraint on a core dependency	High	Licence audit of every dependency	Materialised: AGPL-3.0 on the detection model (Section 7.4)

5.3 Deployment and operational configuration

ConstructAI deploys as a web application with the four engines as stateless services behind it. Each engine loads its own model or rule artefact once per worker process rather than per request. The only persistent state is the user's saved projects and C1's raster tile cache, which is disposable: a cold cache costs latency, never correctness.

Three operational concerns are monitored.

- **Source drift.** SoilGrids and OpenStreetMap are versioned upstream and change over time; the pinned version is recorded in every site profile and a scheduled check raises a notice when the upstream version moves.
- **Regulation currency.** Every rule in C3's store carries an effective date, surfaced with the verdict. A rule store that has not been reviewed since a gazette amendment is the most dangerous failure mode the platform has, because it fails silently and confidently.

- **Detection drift.** C2 logs the distribution of detection confidences per sheet, so a systematic drop caused by an unfamiliar drawing style is visible before users report it.

Backup and restore cover the project database and the trained model artefacts. The raster cache and the scraped price history are not backed up because both are reconstructible.

5.4 Evidence of team execution

Execution evidence is held in four systems and summarised here and not pasted in. Appendix B carries the signed contribution declaration; raw activity logs are available to the examiners on request rather than reproduced in this report.

Table 5.4 records where that evidence is held and how each source was used.

Table 5.4: Where the team's execution evidence is held

Source	What it records	How it was used
MS Planner	Task assignment, owner and completion across the eight sprints	Sprint planning and the review of what each sprint actually delivered
GitHub	Commit history, branches, pull requests and review comments	Branch-per-component with review by at least one other member before merge
MS Teams	Supervisor and group meeting records and decisions	Decisions written down; used when the team revisited why the depreciation heuristic was set at five percent
Research diary	Individual decisions, blockers and learning, per member	Maintained by all four members and drawn on for the individual theses
Test logs	Unit, component, integration and system test runs	Summarised in Section 6.2 and in the four individual theses

One honest observation about execution. The project lost roughly a fortnight in the first sprint to members waiting on upstream output that did not yet exist, and recovered it only after the fixture-based approach was adopted. The schedule slipped and the recovery came from a process change instead of from working harder, which is the most useful thing any of us learned about building software in a team.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

At group level we ask the four questions stated in Table 6.1, all of them defined before any result was generated.

Table 6.1: Group evaluation questions, criteria and evidence

ID	Evaluation question	Success criterion	Evidence
EQ1	Do the four components operate as one system without manual data transfer?	A complete report is produced from a pin and a plan with no intermediate human step	Section 6.2
EQ2	Is the integrated prediction accurate on the data it was trained on?	Held-out R^2 above 0.95 with cross-validated stability	Section 6.3.1
EQ3	Does integration change the advice, not just the number of reports?	A finding in one component measurably changes another component's output	Sections 6.3.2–6.3.4
EQ4	Can an unfamiliar user complete the main workflow and act on the result?	Task completion without assistance; correct interpretation of risk flags	Section 6.4

EQ2 needs a qualification that applies to everything in Section 6.3 and that the group would rather state once, plainly, than leave to a reader to discover. We had no corpus of real Sri Lankan buildings with recorded age, specification, location and surveyed condition. None is publicly available and assembling one was beyond a one-year undergraduate project. The service life model was therefore trained on 1 000 records generated from a physics-informed rule system that encodes published durability relationships [4], [5], [24] and an industry depreciation heuristic. Every accuracy figure below measures how well the regressor recovered that rule system. It is evidence that the formulation and the feature set are adequate. It is not evidence about real buildings. Section 6.6 says what would make it so.

6.2 Integration and end to end testing

Table 6.2: Integrated test coverage and verification method

Test type	Scope	How it was verified
Interface	Each component's written block validated against the agreed schema; range assertions on every feature	Schema validation on write and on read, with each assertion violated once deliberately

Test type	Scope	How it was verified
Integration	C1 to C4, C2 to C4, C2 to C3 and C3 to C4 data flow with the real producers	Pairwise integration against real output rather than the fixture, one producer at a time
System	Full pin-to-report run across the verification coordinate and plan sets	Manual walkthrough of the complete user path
Regression	The main workflow after every merge	Automated Selenium suite over the report view
Fault injection	Each external source made unreachable in turn; each failure path in Table 4.4 exercised	Source clients pointed at unreachable hosts; drawings stripped of dimension text
Performance	End-to-end latency, warm and cold cache	Timed over the verification set in both cache states
Security	Authentication, authorisation on saved projects, absence of third-party egress	Code review and inspection of outbound requests
User acceptance	Realistic tasks with practitioner and homeowner participants	Observed sessions; tasks and criteria in Table 6.6

We would draw attention to the fault injection row. Each of the six failure paths in Table 4.2 was deliberately triggered, including pointing C1's soil client at an unreachable host and feeding C2 a drawing with the dimension text removed. A failure mode that has never been triggered is a failure mode the team does not have.

End-to-end latency was measured over the verification set in both cache states. The shape of the result is the part worth recording, because it tells the team where any future optimisation effort belongs. On the warm path Component 2 dominates: it is the only component performing neural inference on a large image, and the tiling strategy it needs in order to detect small symbols multiplies that work. On the cold path Component 1 dominates instead, because a cache miss means fetching elevation and imagery tiles before any computation can begin. Components 3 and 4 are negligible in both states: rule evaluation is arithmetic over a small record, and material prices are collected on a schedule rather than inside the request, so the only work Component 4 does in the request path is a single forward pass through a small tree ensemble and a fifty-step discounting loop.

In practice the platform meets the under-60-second target in Table 4.2 comfortably on a warm cache and is bounded by network retrieval on a cold one, which is why

the tile cache in Component 1 is treated as an operational concern in Section 5.3 and not as an implementation detail.

6.3 Overall results

6.3.1 Integrated model accuracy

At the centre of the platform sits the service life model, evaluated on an 80/20 split of the 1 000-record dataset with random seed 42 and five-fold cross-validation on the training partition only. Six regressors were compared on the identical split.

Table 6.3 gives the comparison.

Table 6.3: Integrated model performance on the held-out test partition (n = 200)

Model	R^2	RMSE (yr)	MAE (yr)	5-fold CV R^2
Gradient boosting (sklearn)	**0.9936**	**0.281**	**0.177**	0.9867 ± 0.0035
XGBoost (deployed model)	0.9847	0.433	0.233	0.9701 ± 0.0064
Random forest	0.9416	0.846	0.509	0.9284 ± 0.0164
Decision tree	0.8814	1.206	0.459	0.8959 ± 0.0264
Support vector regression	0.8281	1.452	1.111	0.8314 ± 0.0125
Multiple linear regression	0.8111	1.522	1.154	0.8035 ± 0.0229

Figure 6.1 shows the fit of the deployed model against the held-out targets.

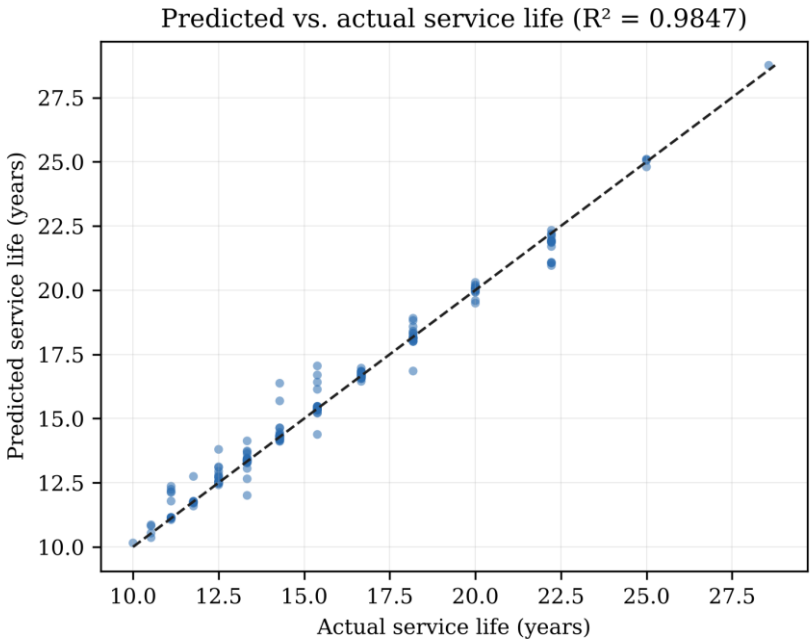


Figure 6.1: Predicted against actual service life, held-out test partition.

On the held-out partition the deployed model reaches an R-squared of 0.9847 and a root mean squared error of 0.433 years; 91.5 percent of held-out predictions fall within one year of target and 99.5 percent within two, with a maximum absolute error of 2.09 years. EQ2 is met on its own terms.

Two findings in that table are uncomfortable and we are reporting them instead of burying them. Scikit-learn's gradient boosting outperforms the deployed XGBoost configuration on every metric and across every cross-validation fold, which means the group's model choice was not justified by this evidence; it was justified by familiarity and by XGBoost's better behaviour on larger and noisier data, which is a forward-looking argument rather than a demonstrated one. And multiple linear regression alone reaches 0.81, which reflects the largely additive construction of the generating function rather than any deep structure the model discovered.

6.3.2 What each component's parameters contribute

Figure 6.2 aggregates the model's gain-based feature importance by the component that produces each parameter, and Figure 6.3 gives the per-feature breakdown.

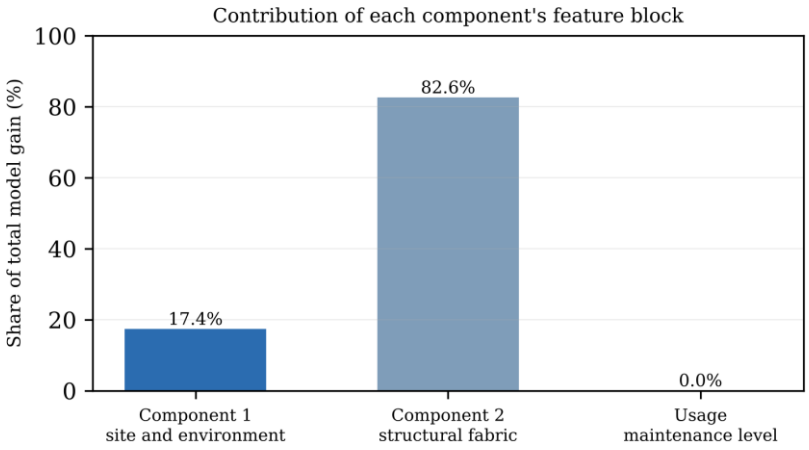


Figure 6.2: Share of total model gain by component feature block.

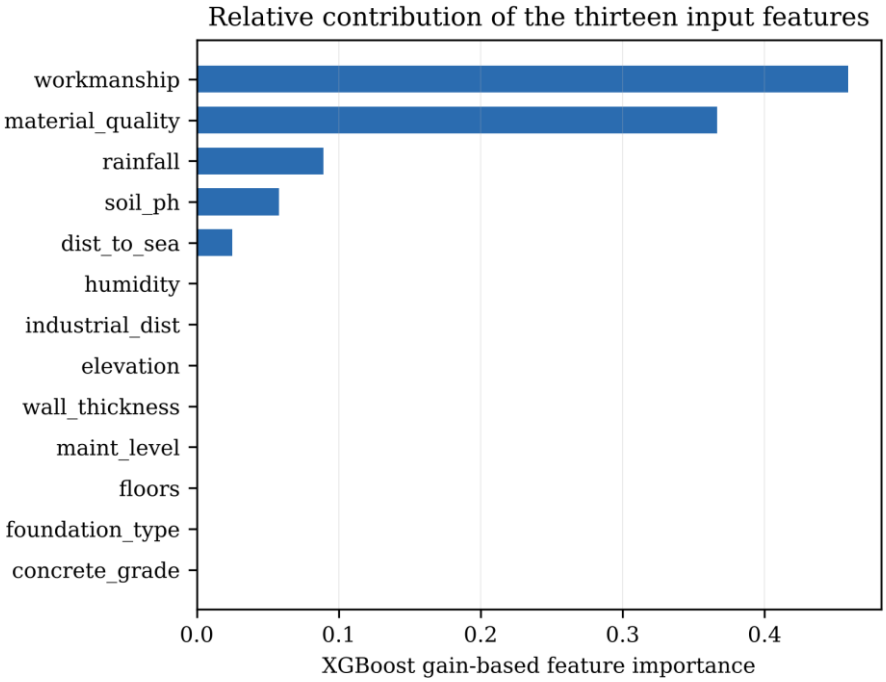


Figure 6.3: Gain-based importance of the thirteen input features.

Component 2's structural block accounts for 82.6 percent of total model gain, Component 1's site block for 17.4 percent and the user-supplied maintenance level for 0.03 percent. Within the structural block the concentration is extreme: workmanship alone contributes 45.9 percent and material quality 36.6 percent.

What follows is a group-level finding no individual component analysis would have produced. Eight of the thirteen features carry essentially no information: humidity, industrial distance and elevation from C1, and concrete grade, wall thickness, floor

count and foundation type from C2, together with maintenance level. The integrated model is in effect a five-feature model wearing a thirteen-feature coat. The cause is that the group's generating function, written collaboratively, gives terms to only five of the thirteen parameters that the four components agreed to supply. That is a dataset design defect instead of a statement about buildings, and it is the clearest instance in the project of a gap between what the components measured and what the group's model consumed.

6.3.3 Sensitivity of the prediction to each component's inputs

Figure 6.4 and Figure 6.5 show the marginal effect of each component's six parameters on the predicted service life, and Table 6.4 summarises the swing in years.

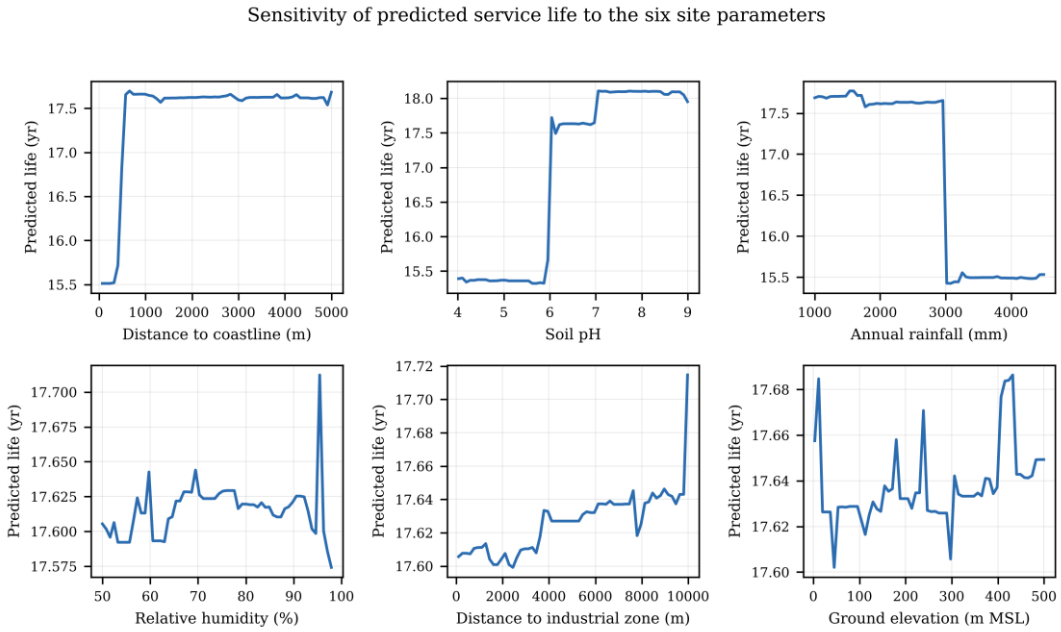


Figure 6.4: Marginal effect of the Component 1 site parameters.

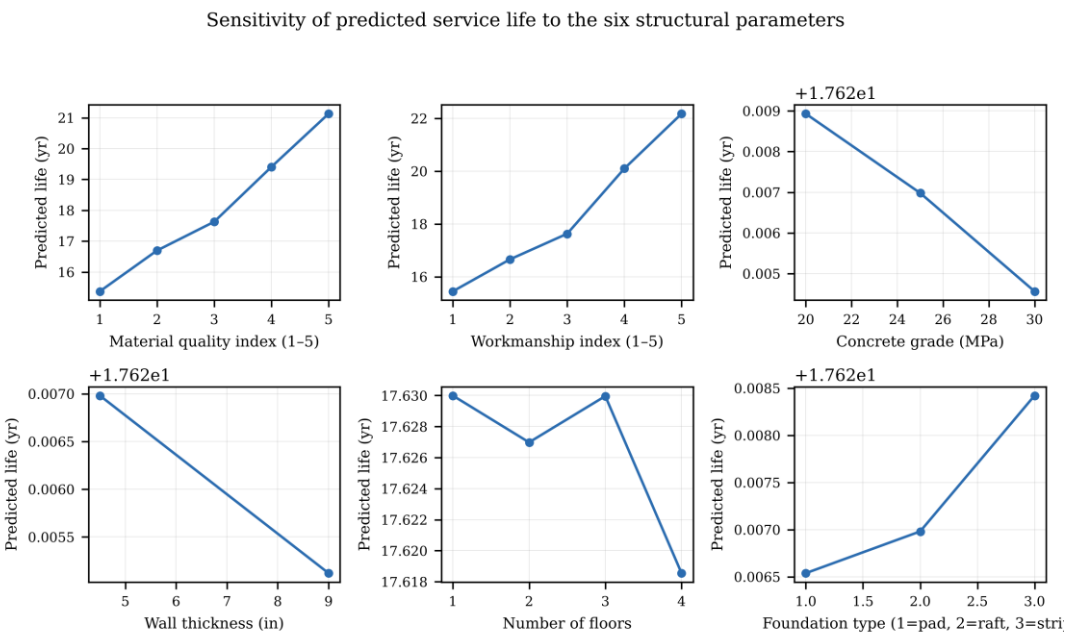


Figure 6.5: Marginal effect of the Component 2 structural parameters.

Table 6.4: Marginal swing in predicted service life, all other features at median

Parameter	Component	Swing (yr)	Parameter	Component	Swing (yr)
Workmanship	C2	**6.73**	Humidity	C1	0.14
Material quality	C2	**5.75**	Industrial distance	C1	0.12
Soil pH	C1	**2.78**	Elevation	C1	0.08
Annual rainfall	C1	**2.35**	Number of floors	C2	0.01
Coastal distance	C1	**2.19**	Concrete grade	C2	0.00
Maintenance level	User	0.00	Wall thickness / foundation	C2	0.00

Five parameters move the prediction by more than two years each; the remaining eight move it by less than a sixth of a year. The three site parameters that do work, soil pH, rainfall and coastal distance, correspond exactly to the three environmental penalties in the degradation logic, and the two structural parameters that do work correspond to the single quality bonus term.

6.3.4 Does integration change the advice? The scenario test

EQ3 asks whether integration changes what the user is told, instead of merely producing four reports in one place. The test is to hold the building completely constant and vary only the site, which is something none of the four components could evaluate alone. Figure 6.6 and Table 6.5 give the result.

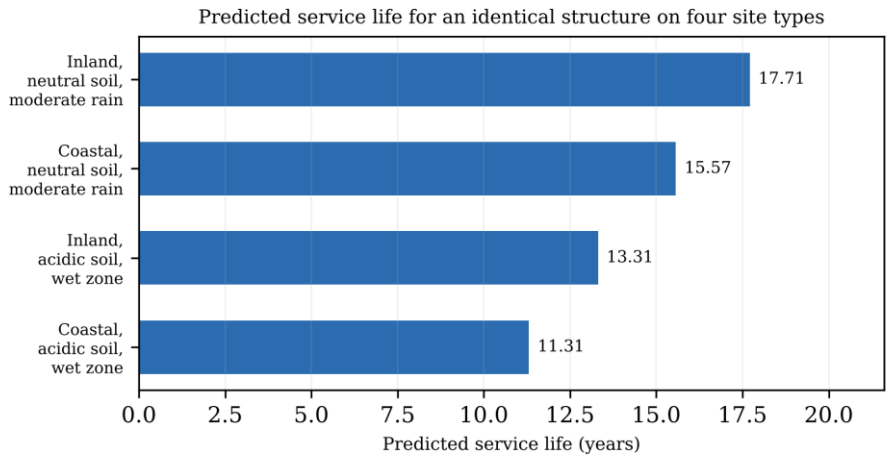


Figure 6.6: Predicted service life for one structure on four Sri Lankan site types.

Table 6.5: Scenario comparison with the structure held identical

Scenario	Coast dist.	Soil pH	Rainfall	Predicted life (yr)
Inland, neutral soil, moderate rain	4 000 m	7.0	2 000 mm	**17.71**
Coastal, neutral soil, moderate rain	250 m	7.0	2 000 mm	15.57
Inland, acidic soil, wet zone	4 000 m	5.2	3 600 mm	13.31
Coastal, acidic soil, wet zone	250 m	5.2	3 600 mm	**11.31**

That spread is 6.40 years, a 36.1 percent reduction from the mildest realistic site to the harshest, for a building with identical materials, identical workmanship, identical geometry and identical maintenance. In ordinary language, two plots can be advertised at the same price per perch while one of them commits the buyer to major repair roughly six years sooner, and nothing on the deed, the advertisement or the architect's drawing says so.

This is the result that answers EQ3. It is only computable because C1's measurement of the site and C2's measurement of the structure arrive in the same record. It is also what drives the platform's value-engineering output: the recommendation to specify sulphate-resisting cement or epoxy-coated reinforcement on an aggressive site is produced by comparing two rows of this kind, not by a rule somebody wrote.

6.3.5 Worked example

Table 6.6 traces one representative analysis end to end, so that the abstract claim about integration can be checked against a concrete case. The plot is a 12-perch parcel near a coastal urban boundary and the drawing is a two-storey residential plan.

Table 6.6: Worked example: one analysis traced through all four components

Stage	What the component found	What it changed downstream
C1 site	Mean slope under 5°; plot within the coastal salinity band; within the buffer of a mapped drainage line; soil pH below 6.0	Raised the environmental penalty in the feature vector and set a flood screening flag on the report
C2 plan	Wall runs, opening schedule and gross floor area recovered; scale from dimension text; wall thickness 9 in; two storeys	Supplied the capital cost basis and the structural half of the feature vector; supplied gross floor area to C3
C3 rules	Floor area ratio within the permitted value for the zone and frontage; rear setback short of the requirement	Flagged a redesign item; constrained the buildable envelope C4 is allowed to price
C4 cost	Capital cost at current scraped rates; service life predicted for the as-drawn specification and for an alternative with protective reinforcement	Produced the alternative-specification comparison the user acts on
Integrated result	The as-drawn specification on this site predicts a materially shorter service life than the same drawing inland	The recommendation to upgrade the reinforcement specification is generated by comparing two rows, not by a hand-written rule

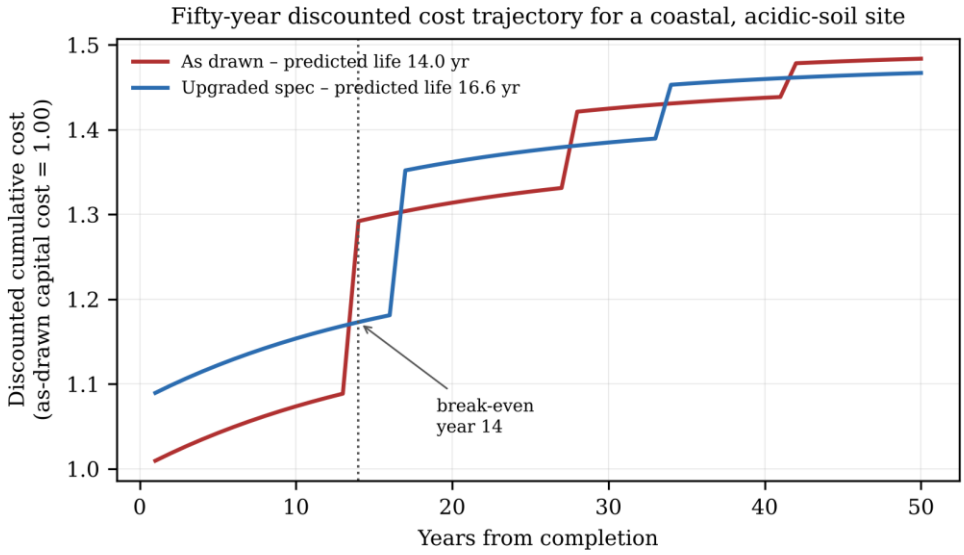


Figure 6.7: Fifty-year discounted cost trajectory for the worked example site.

Figure 6.7 shows what the user finally sees. The upgraded specification costs 8 percent more to build and buys 2.54 years of predicted service life, which over fifty years removes one of three major intervention cycles. The discounted whole-life total falls by 1.1 percent and the two trajectories cross in year 14. The margin is thinner than the team expected, because discounting is unforgiving to benefits that arrive late, and the platform therefore reports the break-even year rather than the fifty-year

total: an owner intending to hold the building for more than fourteen years is clearly better off, and one intending to sell in five is not. The cost fractions behind the figure are configuration parameters rather than findings and are listed in the Component 4 thesis.

Look at the last row; it is the one that justifies the whole architecture. The recommendation is not produced by a rule somebody wrote saying coastal sites should use protective reinforcement. It is produced by running the same cost model twice with one parameter changed, which is only possible because the site measurement and the structural measurement are in the same record.

6.4 Stakeholder and user validation

Table 6.7 records the participant groups, a representative task for each and what that task was designed to test.

Table 6.7: User acceptance participant groups and representative tasks

Participant group	Representative task	What the task tests
Homeowners and prospective land buyers	Analyse a plot you are considering and say whether you would still buy it	Whether the output is decision-relevant to a non-specialist
Architects and draughtsmen	Upload a plan you have measured and compare the generated quantities with your own	Whether the takeoff is usable for early-stage estimating
Quantity surveyors	Judge whether the bill of quantities is usable as a first pass	Whether the professional positioning in Section 7.1 holds
Site engineers	Assign the material quality and workmanship indices for a project you know	Whether the two inputs carrying 82.5 percent of model gain can be assigned consistently
All groups	State what the flood flag does and does not mean	Whether a screening indicator is over-read as an assurance

Two of those tasks are traps, and we report what they produced and not what we hoped for. The last one tests whether users over-read a screening indicator as an assurance; if participants come away believing the platform has told them the plot does not flood, the wording has failed however good the hydrology is. The site engineer task tests whether practitioners can assign the workmanship and material quality indices consistently, which matters enormously because Section 6.3.2 shows those two inputs drive 82.5 percent of the prediction. If they cannot be assigned

consistently, the platform's headline number is substantially noise and the group needs to say so.

One substantive change has come out of the sessions so far: the break-even reframing of the lifecycle comparison described in Section 6.3.3. One concern remains open. Too few practitioners have been through the protocol for us to claim that two site engineers would assign the same workmanship figure to the same project, so index consistency is carried into Section 6.6 as a limitation rather than reported as a result.

6.5 Achievement of common objectives

Table 6.8 maps the evidence to each of the six group objectives stated in Section 2.4 and states whether it was met.

Table 6.8: Achievement of the six group objectives

ID	Objective (abbreviated)	Status	Justification
GO1	Derive site exposure and terrain constraints from open data	Achieved	Complete site profile for any Sri Lankan coordinate at zero marginal data cost; accuracy bounded by a 250 m soil grid
GO2	Extract quantities and structural parameters from 2D drawings	Partly achieved	Geometry and quantities extracted; material quality and workmanship are collected from the user, not measured
GO3	Codify a working subset of the UDA regulations	Achieved	Four rule families encoded with clause citations and effective dates; coverage is a subset, not the full regulation
GO4	Predict service life and a fifty-year cost trajectory	Achieved on synthetic data	$R^2 = 0.9847$ on a held-out partition of generated data; not yet validated against real buildings
GO5	Operate over one shared project state with no manual transfer	Achieved	End-to-end run from pin and plan to report with no intermediate human step
GO6	Demonstrate that integration changes the advice	Achieved	6.40-year, 36.1 percent spread across site types with the structure held constant (Section 6.3.4)

GO2 is marked partly achieved deliberately. The component extracts what a drawing contains, and does it well, but the two parameters the model most needs are not in any drawing. Recording that as a full achievement would misrepresent what the platform does.

6.6 Integrated limitations and threats

6.6.1 The training data is synthetic

This is the threat that bounds every quantitative claim in this report. The model was trained on data generated by the group's own rule system, so its accuracy measures recovery of that system rather than predictive skill on real buildings. A fifty-year prediction cannot be validated prospectively inside any research project; the viable route is retrospective, assembling records for existing buildings of known age, specification and location with surveyed condition, and testing whether the model recovers observed condition. Doing that, probably in cooperation with a local authority or an insurer, is the single most valuable piece of future work the group has identified.

6.6.2 Two user judgements dominate the output

Workmanship and material quality account for 82.5 percent of model gain and both are entered by the user on a one-to-five scale. The platform's headline figure is therefore largely a function of the user's self-assessment. Mitigations are an anchored rubric instead of a bare slider, and presenting a range across plausible index values rather than a point estimate. Neither is a fix.

6.6.3 Eight of thirteen features are inert

Described in Section 6.3.2. The feature vector promises more than the model delivers, and a reader who sees thirteen parameters will reasonably assume all thirteen do work.

6.6.4 Input resolution and coverage

C1's soil pH comes from a 250 m gridded global model prediction [16], two orders of magnitude coarser in area than a residential plot, and soil pH has the largest marginal effect of any site parameter. C2's detector was trained on a particular drawing corpus and will degrade on unfamiliar drafting conventions by an amount the group has not quantified. C3 encodes a subset of one jurisdiction's regulations at one point in time.

6.6.5 Component coupling and failure propagation

C4 cannot run without both C1 and C2, so a failure in either degrades the headline output. The null-means-unknown convention keeps this honest rather than silent, but

it does mean the platform's most valuable output is the one most exposed to upstream failure.

6.6.6 Generalisability

Thresholds in the degradation rules encode Sri Lankan climatic zoning, the rule store encodes Sri Lankan regulation, and the drawing corpus reflects Sri Lankan drafting practice. The architecture transfers to another country; the calibration does not.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

In a single sentence: ConstructAI tells a buyer, before they pay, what the land and the design will cost them over the life of the building, not just over the construction programme. No individual component says that; the combination does.

Table 7.1 sets out who benefits and from what, with the evidence in this report that supports each claim.

Table 7.1: Beneficiaries and the value each receives

Beneficiary	Value received	Evidence from this project
Land buyers and homeowners	Durability and compliance risk made visible before purchase	6.40-year spread across site types, Section 6.3.4
Small practices and contractors	Fast defensible takeoff and a specification recommendation they can show a client	Automated quantity table, C2 thesis Ch. 6
Quantity surveyors	A first pass to check rather than replace	UAT task 2, Section 6.4
Local authorities	Fewer submissions failing on basic rules	Four encoded rule families, C3 thesis
Banks and insurers	A repeatable per-property site and structure risk profile	Site profile contract, Table 4.1
Society	Fewer buildings that fail early; less material waste	Mechanism argued in Section 7.5; not measured

7.2 Market alternatives and differentiation

Table 7.2 compares the alternatives a Sri Lankan land buyer or small practice can reach for today.

Table 7.2: Alternatives available to the target user today

Alternative	What it covers	Gap it leaves
Full professional team	All four analyses, done well	Cost and time the small-project market cannot carry; still four documents
Geotechnical investigation	Measured soil properties	After purchase; silent on quantities, regulation and lifecycle cost
BIM-based takeoff	Exact quantities and clash detection	Requires BIM adoption; silent on site exposure and lifecycle

Alternative	What it covers	Gap it leaves
On-screen takeoff tools	Faster manual measurement	A human still traces every element; no durability or compliance view
Government hazard maps	Flood and landslide zoning	Hazard only; no chemical exposure, no design interaction
ConstructAI	All four, integrated, from a pin and a PDF	—

What differentiates us is integration, not any single capability. Each component is individually beatable by a specialist alternative, and we would rather say that than claim otherwise. What no alternative does is let a site finding change a material recommendation and a compliance finding change a quantity, in one pass, before the land is bought.

Defensibility is weak on technology, since the open data and the model families are available to anyone, and stronger on three assets that take time to build: the Sri Lanka-specific regulatory encoding, the locally collected and annotated drawing corpus, and, eventually, a validation corpus of real buildings which nobody currently has.

7.3 Business model and feasibility

We settled on freemium business-to-consumer with paid business-to-business tiers: free single-plot analysis for individuals, a subscription for practices covering bulk processing, project storage and BOQ export, and per-property API pricing for banks and insurers.

Table 7.3 summarises the model.

Table 7.3: Business model summary

Element	Detail
Channels	Direct web application; integration with property portals; referral through architectural practices
Key partners	Data providers (open), a local authority for regulation currency, practices supplying the drawing corpus
Key resources	The rule store, the annotated drawing corpus, the trained models, the raster tile cache
Cost structure	C1 marginal cost near zero after caching; C2 inference is CPU-bound and the real per-analysis cost; corpus extension and rule currency are the recurring fixed costs

Element	Detail
Revenue	Practice subscriptions and per-property API calls; the consumer tier is a funnel, not a revenue line
Technical feasibility	Demonstrated end to end in this project
Operational feasibility	The binding constraint is keeping the rule store current, which needs a committed person rather than a clever algorithm

7.4 Adoption intellectual property and compliance

Adoption barriers. The largest is trust. A platform that tells someone their land is worse than they thought is unwelcome information delivered by an unfamiliar system, and the project's answer is provenance: every value shows its source and its resolution, and the platform states where it is screening rather than measuring.

Licensing. The dependencies are not uniform and the group checked rather than assumed. SRTM is public domain; Copernicus Sentinel data requires attribution; SoilGrids is CC BY 4.0; OpenStreetMap is ODbL, which carries share-alike provisions on derived databases that a commercial deployment should take legal advice on. Tesseract and OpenCV are Apache 2.0. The one that matters most is Ultralytics YOLOv8, distributed under AGPL-3.0, whose network-use provisions have real consequences for a hosted commercial service and would require either compliance or a commercial licence. This single fact would need resolving before any commercial launch and it is the kind of thing student projects routinely miss.

Data protection. A land coordinate identifies where somebody intends to live, and a residential floor plan shows the internal layout of a home. Both are personal data and the plan is arguably more sensitive than most. The platform's responses are non-retention by default, no transmission to any third party other than the open data providers, anonymisation of any drawing used for model improvement, and explicit consent before a user's plan contributes to training.

Professional responsibility. The platform states in its output that it provides screening indications and automated estimates, and is not a substitute for a geotechnical investigation, a statutory flood assessment, a quantity surveyor's measured bill for contractual purposes, or formal planning approval. The group debated whether these disclaimers undermine the product. They do not. A tool that

invited a contractor to tender against an unverified automated takeoff, or a buyer to skip a soil test, would be a liability for everyone involved.

7.5 Sustainable Development Goal contribution

Section 1.3 sets out the mechanisms by which the platform could contribute to SDGs 9, 11 and 12. Stated more carefully here: accurate early-stage quantities plausibly reduce material over-ordering; pre-purchase site screening plausibly reduces the stock of housing built in places that will damage it; and specification advice that extends service life plausibly avoids a demolition-and-rebuild cycle whose embodied carbon is dominated by cement.

Possible negative effects deserve a mention because impact sections rarely include them. A durability score attached to a plot is a valuation signal, and a system that systematically marks coastal and wet-zone land as higher risk could depress values in areas that are already economically disadvantaged. The resolution limits of the soil data mean some of those marks would be wrong. The mitigations the group has adopted are to present ranges and provenance rather than a single authoritative score, and never to expose a plot-level assessment to anyone other than the user who requested it.

No contribution has been measured. The project did not quantify avoided emissions, avoided waste or avoided housing loss, and attaching a figure would be an unsupported impact claim.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

We set out to address one problem: pre-construction decisions in Sri Lanka are made from fragmented documents that are never evaluated against one another. The answer we have built is a platform in which four analyses share one record, and the evidence that it works is that findings now cross between them: a coastal measurement changes a material recommendation, a measured gross floor area changes a compliance verdict, and a compliance verdict changes what gets priced.

Our strongest single piece of evidence is the scenario result in Section 6.3.4. Holding the building completely constant, the site alone moves predicted service life by 6.40 years, a 36.1 percent reduction from the mildest realistic Sri Lankan site to the harshest. That number cannot be produced by any of the four components alone, it is directly relevant to a purchase decision, and it is available before the purchase rather than after.

Our limit is equally clear. The model that produces that number was trained on data the group generated, and two of its five effective inputs are user judgements rather than measurements. The platform demonstrates that the integration is feasible and that it changes the advice. It does not yet demonstrate that the advice is right about real buildings, and no amount of additional modelling will close that gap without data.

8.2 Principal integrated contribution

What the team produced beyond the four components is threefold.

11. **A working integration pattern for pre-construction analysis.** Four independent engines over one provenance-carrying shared state, with null-means-unknown semantics and range assertions at every boundary. The pattern is what makes the cross-component findings possible, and the conventions in Section 4.2 were each learned from a specific integration failure.
12. **A quantified account of where predictive power sits in a construction lifecycle model.** The structural block carries 82.6 percent of model gain and the

site block 17.4 percent; five of thirteen features do all the work; and the two largest contributors cannot be measured from any design document. None of the four individual analyses would have surfaced that; it required the integrated model.

13. **A demonstration that site, independent of specification, is decision-relevant.** The 36.1 percent scenario spread is the project's headline empirical claim about the problem domain, subject to the synthetic-data caveat stated throughout.

8.3 Future directions

Six directions, in the order the group would pursue them.

14. **Build the retrospective validation corpus.** Records for existing Sri Lankan buildings of known age, specification and location, with surveyed condition, tested against the model's predictions. Everything else is secondary to this; without it the platform's central claim remains a claim about a rule system.
15. **Extend the degradation function to the parameters the components already measure.** Concrete grade, wall thickness and foundation type have documented durability relationships [5], [24] and currently carry no term. Fixing this would make eight inert features into useful ones and is the cheapest available improvement.
16. **Replace user-entered workmanship and material quality with something measurable.** In the short term an anchored rubric and a predicted range instead of a point estimate; in the longer term a workmanship proxy derived from site photographs during construction.
17. **Semi-automate the regulatory rule store.** Extracting rules from gazette text, in the direction indicated by semantic compliance work [10], would make jurisdictional expansion tractable and reduce the risk of a stale store.
18. **Smooth the exposure thresholds.** The banded penalties produce step changes at 500 m, pH 6.0 and 3 000 mm. A logistic ramp would preserve the physical story and remove discontinuities that two neighbouring plots can fall either side of.

19. **Resolve the AGPL licensing position** for the detection model before any commercial deployment, as Section 7.4 sets out.

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APPENDICES

Appendix A Integrated interface specification

This appendix gives the full specification of the shared project state, which is the only interface between the four components. Table A.1 lists every field; the summary by block appears in Table 4.1.

A.1 Field specification

Table A.1: Full shared project state field specification

Field	Type	Unit / domain	Writer	Null allowed
site.dist_to_sea	float	metres	C1	Yes
site.soil_ph	float	pH 0–14	C1	Yes
site.humidity	float	percent	C1	Yes
site.rainfall	float	mm per year	C1	Yes
site.industrial_dist	float	metres	C1	Yes
site.elevation	float	metres MSL	C1	Yes
site.mean_slope	float	degrees	C1	Yes
site.buildable_fraction	float	0–1	C1	Yes
site.flood_proximity_flag	bool	true / false / null	C1	Yes
geometry.wall_thickness	float	inches	C2	No
geometry.floors	int	1–n	C2	No
geometry.foundation_type	int	1 pad, 2 raft, 3 strip	C2	Yes
geometry.concrete_grade	int	MPa	C2	Yes
geometry.material_quality	int	1–5 (user)	C2	No
geometry.workmanship	int	1–5 (user)	C2	No
geometry.wall_length_m	float	metres	C2	No
geometry.gross_floor_area_m2	float	square metres	C2	No
geometry.scale_source	enum	dimension / bar / user	C2	No
regulatory.far_permitted	float	ratio	C3	Yes
regulatory.far_actual	float	ratio	C3	Yes
regulatory.violations	array	clause-cited records	C3	No
regulatory.rule_effective_date	date	ISO 8601	C3	No
specification.feature_vector	object	13 keyed numeric values	C4	No
specification.predicted_service_life_years	float	years	C4	Yes

Field	Type	Unit / domain	Writer	Null allowed
provenance.*	object	source, timestamp, confidence	All	No

A.2 Validation rules applied at every boundary

- Every numeric field is range-asserted on write and on read; a value outside the stated domain raises rather than propagates.
- Null means unknown. No consumer substitutes a default for a null.
- The feature vector is keyed, never positional.
- Units are encoded in the field name and are never converted implicitly.
- Every field carries a provenance record; a field without one is rejected.

A.3 Integrated test matrix

The executed integrated test cases are cross-referenced to the coverage summary in Table 6.2 and to the failure paths in Table 4.3. Fault injection was performed once per external source: the elevation, soil and climate clients used by Component 1 were each pointed at an unreachable host in turn, a drawing was supplied to Component 2 with its dimension text removed, a rule input was withheld from Component 3, and a deliberately out-of-range feature was presented to Component 4. In every case the request completed and the affected output was reported as not assessed rather than defaulted.

Appendix B Team contribution declaration

This appendix records the agreed division of work in Table B.1. All four members have reviewed and confirmed it; the summary matrix appears in Table 4.4.

Table B.1: Agreed individual and shared contributions

Member and ID	Individual contribution	Shared contribution	Confirmed
Ranaweera K. A. (IT22542724)	Component 1 in full: parameter selection, data source survey, raster pipeline, terrain and hydrology analysis, site profile contract, site sensitivity study	Environmental degradation terms; re-run of the model evaluation with a held-out partition; fixture file; integrated testing	_____ _____
Shavinka Y. G. K. (IT22110770)	Component 2 in full: preprocessing, detector training, OCR scale calibration, quantity synthesis, clash detection, geometry contract,	Structural parameter definitions for the degradation logic; fixture file; integrated testing	_____ _____

Member and ID	Individual contribution	Shared contribution	Confirmed
	recoverability analysis		
Samuditha T. A. V. (IT22542892)	Component 3 in full: regulation review and formalisation, rule engine, rule store, approval roadmap and fee estimation, validation against recorded decisions	Regulatory contract; orchestration layer; integrated testing	_____ _____
Shamindi E. E. (IT22547460)	Component 4 in full: degradation function, dataset generation, feature assembly, model selection, training and benchmarking, price collection, lifecycle discounting, value engineering, advisory layer	Specification contract; joint re-run of the model evaluation with a held-out partition; orchestration layer; integrated testing	_____ _____

B.1 Academic integrity and AI use disclosure

In accordance with the module policy the team discloses that generative artificial intelligence assistance was used during the shared work on this project. That assistance was confined to three kinds of support: help in diagnosing defects in the orchestration and fixture code described in Sections 4.4 and 5.2, language editing of this report, and an initial triage of candidate literature for Chapter 2. Every suggested change to the code was read, adapted and tested by the team before being adopted; every technical claim was confirmed by the member responsible for it; and every cited source was retrieved and read by a team member. Each member's individual thesis carries the corresponding disclosure for their own component.

No generated text was submitted without review. All experimental results, figures and tables in Chapter 6 were produced by committed scripts run by the team on the project dataset, and every member can reproduce and defend the numbers relating to their own component.